# LAB ASSIGNMENT – X

Q1. Insert a new customer and open an account.

```
himansh1241 — mysql -u root — 80×8

mysql> INSERT INTO Customers VALUES (104, 'Rohit', 'Chennai', '9871234567');
Query OK, 1 row affected (0.001 sec)

mysql>
mysql> INSERT INTO Accounts VALUES (2004, 104, 'Savings', 12000);
Query OK, 1 row affected (0.001 sec)

mysql>
```

Q2. Deposit ₹2000 into account 2001.

```
himansh1241 — mysql -u root — 85×12

mysql> UPDATE Accounts
    -> SET Balance = Balance + 2000
    -> WHERE Acc_no = 2001;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO Transactions VALUES (9003, 2001, 'Deposit', 2000, CURRENT_DATE);
Query OK, 1 row affected (0.001 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.000 sec)
```

Q3. Withdraw ₹1000 from account 2001 and rollback.

```
himansh1241 — mysql -u root — 85×12

mysql> UPDATE Accounts
    -> SET Balance = Balance - 1000
    -> WHERE Acc_no = 2001;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO Transactions VALUES (9004, 2001, 'Withdraw', 1000, CURRENT_DATE);
Query OK, 1 row affected (0.001 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.000 sec)
```

Q4. Transfer ₹3000 from Acc_no 2001 to Acc_no 2002.

```
himansh1241 — mysql -u root — 85×19

mysql> UPDATE Accounts
    -> SET Balance = Balance - 3000
    -> WHERE Acc_no = 2001;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Accounts
    -> SET Balance = Balance + 3000
    -> WHERE Acc_no = 2002;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO Transactions VALUES (9005, 2001, 'Transfer', 3000, CURRENT_DATE);
Query OK, 1 row affected (0.001 sec)

mysql> INSERT INTO Transactions VALUES (9006, 2002, 'Transfer', 3000, CURRENT_DATE);
Query OK, 1 row affected (0.001 sec)

mysql> COMMIT;
```

Q5. Delete a customer (Cust_id=101), create a savepoint, and rollback.

```
himansh1241 — mysql -u root — 91×16

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.000 sec)

mysql> DELETE FROM Customers
    -> WHERE Cust_id = 101;
Query OK, 1 row affected (0.001 sec)

mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.001 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.000 sec)
```

Q6. Update account type of Acc_no=2002 to “Savings” and commit.

himansh1241 — mysql -u root — 91×11

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql> UPDATE Accounts
    -> SET Acc_type = 'Savings'
    -> WHERE Acc_no = 2002;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.001 sec)
```

Q7. Insert two new customers with savepoints, rollback second only.

himansh1241 — mysql -u root — 91×17

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql> INSERT INTO Customers VALUES (105, 'Anjali', 'Pune', '9998887776');
Query OK, 1 row affected (0.000 sec)

mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.000 sec)

mysql> INSERT INTO Customers VALUES (106, 'Karan', 'Jaipur', '8887776665');
Query OK, 1 row affected (0.000 sec)

mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.000 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.001 sec)
```

Q8. Deduct ₹500 from all savings accounts and rollback.

himansh1241 — mysql -u root — 91×11

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql> UPDATE Accounts
    -> SET Balance = Balance - 500
    -> WHERE Acc_type = 'Savings';
Query OK, 4 rows affected (0.001 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql> ROLLBACK;
Query OK, 0 rows affected (0.001 sec)
```

Q9. Insert a new transaction record without committing, then rollback.

```
himansh1241 — mysql -u root — 91×10
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql> INSERT INTO Transactions VALUES (9007, 2001, 'Deposit', 1500, CURRENT_DATE);
Query OK, 1 row affected (0.001 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.001 sec)

mysql>
```

Q10. Delete all accounts with balance < 5000, then rollback.

```
himansh1241 — mysql -u root — 91×11
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql> DELETE FROM Accounts
    -> WHERE Balance < 5000;
Query OK, 0 rows affected (0.001 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.000 sec)

mysql>
```